



# Statistics of Financial Markets

## Chapter 3: The Efficient Market Hypothesis



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## Learning Objectives

By the end of this chapter, you will be able to:

1. **Define** market efficiency and its three forms (weak, semi-strong, strong)
2. **Formalize** the EMH using martingale and random walk models
3. **Implement** statistical tests for weak-form efficiency (autocorrelation, runs, variance ratio)
4. **Analyze** event studies for semi-strong efficiency
5. **Evaluate** market anomalies and behavioral challenges to EMH
6. **Discuss** the Adaptive Market Hypothesis as a modern synthesis

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## Can You Beat the Market?

- ▣ **Central question in finance:** can an investor consistently earn returns above the market benchmark?
- ▣ Burton Malkiel (1973): “A blindfolded monkey throwing darts at a newspaper’s financial pages could select a portfolio that would do just as well as one carefully selected by experts.”
- ▣ **Empirical fact:** over 90% of actively managed funds underperform their benchmark index over 15-year horizons (SPIVA Scorecard, 2024)
- ▣ If prices already reflect all available information, then no strategy can systematically generate **alpha**



## Historical Context

| Year | Author          | Contribution                                                    |
|------|-----------------|-----------------------------------------------------------------|
| 1900 | Louis Bachelier | Thesis: stock prices follow a random walk                       |
| 1953 | Maurice Kendall | UK stock prices exhibit no predictable pattern                  |
| 1965 | Eugene Fama     | "Random Walks in Stock Market Prices"                           |
| 1965 | Paul Samuelson  | "Proof That Properly Anticipated Prices Fluctuate Randomly"     |
| 1970 | Eugene Fama     | Formal EMH taxonomy: weak, semi-strong, strong                  |
| 1991 | Eugene Fama     | Revised EMH: return predictability, event studies, private info |
| 2004 | Andrew Lo       | Adaptive Market Hypothesis (AMH)                                |
| 2013 | Fama & Shiller  | Nobel Prize (opposing views on efficiency)                      |

## Definition of Market Efficiency

### Definition 1 (Efficient Market — Fama, 1970)

A market is **informationally efficient** if prices at any time  $t$  fully reflect all available information  $\Omega_t$ . Formally:

$$P_t = \mathbb{E}[P_t \mid \Omega_t]$$

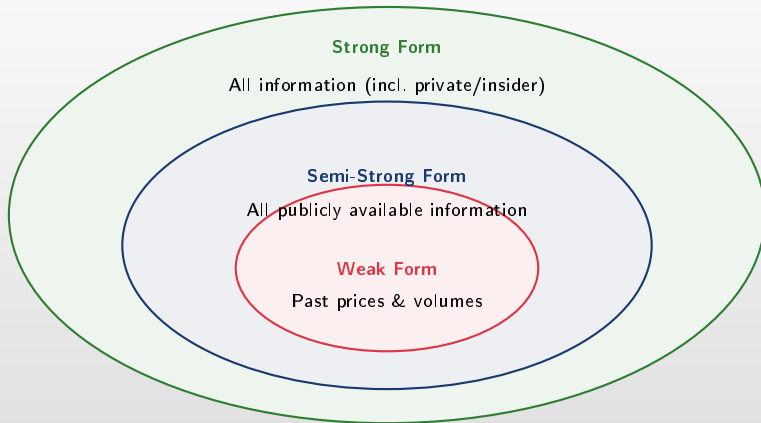
where  $\Omega_t$  is the information set available at time  $t$ .

- **Implication:** no trading strategy based on  $\Omega_t$  can yield expected returns exceeding the equilibrium return
- Price changes respond only to **new information** (surprises), which is by definition unpredictable
- The EMH is a **joint hypothesis**: efficiency + an equilibrium model of expected returns

### Joint Hypothesis Problem

- We can never test efficiency alone — any test requires an asset pricing model (CAPM, APT, etc.). A rejection might mean inefficiency *or* a wrong model.

## Three Forms of the EMH



## The Three Forms: Summary

| Form        | Information set          | Implication                          | What it rules out                 |
|-------------|--------------------------|--------------------------------------|-----------------------------------|
| Weak        | Past prices, volumes     | Tech. analysis $\nRightarrow \alpha$ | Moving averages, chart patterns   |
| Semi-strong | All public info          | Fund. analysis $\nRightarrow \alpha$ | Earnings strategies, news trading |
| Strong      | All info (incl. insider) | No investor has edge                 | Insider trading, private tips     |

- Each form is **nested**: strong  $\Rightarrow$  semi-strong  $\Rightarrow$  weak
- If the strong form holds, both weaker forms hold automatically
- Most empirical research focuses on weak and semi-strong efficiency



## Fair Game Model

### Definition 2 (Fair Game)

Let  $r_{t+1}$  be the return on an asset. The market is a **fair game** with respect to information  $\Omega_t$  if:

$$\mathbb{E}[r_{t+1} - \mathbb{E}[r_{t+1} \mid \Omega_t] \mid \Omega_t] = 0$$

- ▣ Define excess return:  $z_{t+1} = r_{t+1} - \mathbb{E}[r_{t+1} \mid \Omega_t]$
- ▣ Fair game condition:  $\mathbb{E}[z_{t+1} \mid \Omega_t] = 0$  — forecast errors are unbiased
- ▣ No trading rule  $\delta(\Omega_t)$  can produce positive expected excess profit:

$$\mathbb{E}[\delta(\Omega_t) \cdot z_{t+1}] = 0$$

### Intuition

- ▣ A fair game does not mean returns are zero — it means **unexpected** returns are zero on average. Investors earn the equilibrium risk premium, but cannot consistently do better.

## Martingale and Random Walk

### Martingale

- $\{P_t\}$  is a martingale w.r.t.  $\Omega_t$  if:

$$\mathbb{E}[P_{t+1} \mid \Omega_t] = P_t$$

- Prices have no predictable drift
- Weaker than random walk: allows for time-varying variance

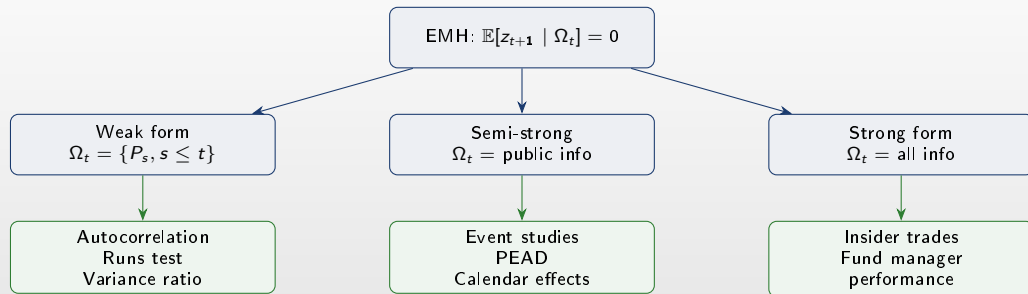
### Random Walk

- **RW1:**  $P_t = P_{t-1} + \varepsilon_t$ ,  $\varepsilon_t \stackrel{iid}{\sim} (0, \sigma^2)$
- **RW2:** increments are independent but not identically distributed
- **RW3:** increments are uncorrelated (weakest)
- $\text{RW1} \Rightarrow \text{RW2} \Rightarrow \text{RW3}$

### Key Distinction

- EMH  $\nRightarrow$  Random Walk. EMH implies a **martingale** for risk-adjusted prices, not necessarily iid increments. Volatility clustering is compatible with EMH.

## From EMH to Testable Predictions



## Testing Weak-Form Efficiency

- ▣ **Null hypothesis:** past returns contain no information about future returns
- ▣ If weak-form EMH holds, returns should be serially uncorrelated:

$$\text{Corr}(r_t, r_{t-k}) = 0, \quad \forall k \geq 1$$

### Main Statistical Tests

| Test            | Null hypothesis                                   | Method                   |
|-----------------|---------------------------------------------------|--------------------------|
| Autocorrelation | $\rho_k = 0$ for all $k$                          | Ljung-Box $Q$ -statistic |
| Runs test       | Returns are random                                | Non-parametric sign test |
| Variance ratio  | $\text{Var}(r_t^{(q)}) = q \cdot \text{Var}(r_t)$ | Lo-MacKinlay (1988)      |
| ADF test        | Unit root in prices                               | Augmented Dickey-Fuller  |

## Autocorrelation Test

### Sample Autocorrelation

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^T (r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^T (r_t - \bar{r})^2}$$

- Under  $H_0$ :  $\hat{\rho}_k \sim \mathcal{N}(0, 1/T)$  approximately
- Reject if  $|\hat{\rho}_k| > z_{\alpha/2}/\sqrt{T}$

### Ljung-Box Q-Statistic

$$Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$$

- Tests joint null:  $\rho_1 = \rho_2 = \dots = \rho_m = 0$
- Under  $H_0$ :  $Q(m) \sim \chi^2(m)$
- Typical choice:  $m = 10$  or  $m = 20$

### Empirical Finding

- Daily returns on major indices show small but statistically significant  $\hat{\rho}_1 \approx 0.05$ – $0.10$  (Lo & MacKinlay, 1988). Economically small, but challenges strict weak-form EMH.

## Python: Autocorrelation and Ljung-Box Test

```
import yfinance as yf
import numpy as np
from statsmodels.stats.diagnostic import acorr_ljungbox
from statsmodels.tsa.stattools import acf

# Download S&P 500 data and compute log-returns
sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
prices = sp['AdjClose'].dropna()
returns = np.log(prices / prices.shift(1)).dropna()

# Sample autocorrelations (first 20 lags)
acf_vals = acf(returns, nlags=20, fft=True)
print("ACF_lags_1-5:", np.round(acf_vals[1:6], 4))

# Ljung-Box test (H0: no autocorrelation up to lag m)
lb = acorr_ljungbox(returns, lags=[10, 20], return_df=True)
print(lb) # If p-value < 0.05, reject weak-form EMH
```

## Autocorrelation Function of S&P 500 Returns

### ACF of Returns $r_t$

- Autocorrelations are very small:  $|\hat{\rho}_k| < 0.05$  for most lags
- A few lags may be marginally significant (barely outside  $\pm 1.96/\sqrt{T}$  bands)
- Consistent with weak-form EMH**: returns are approximately unpredictable

### ACF of $|r_t|$ and $r_t^2$

- Absolute and squared returns show **strong, persistent** autocorrelation
- $\hat{\rho}_k(|r_t|) > 0$  for  $k = 1, \dots, 100+$
- This is **volatility clustering**: compatible with EMH (martingale allows time-varying variance)
- Key distinction: **returns** are unpredictable, but **volatility** is predictable

### Important

- Predictable volatility  $\neq$  predictable returns. EMH is about the **direction** of price changes, not their magnitude.

## Autocorrelation Function: Empirical Results

*[Chart: ACF of S&P 500 returns and squared returns — ch3\_emh\_tests.pdf]*

- ▣ **Left:** ACF of returns — all bars within the 95% confidence band
- ▣ **Right:** ACF of  $r_t^2$  — significant persistent autocorrelation (volatility clustering)

 SFM\_ch3\_emh\_tests



## The Runs Test

### Definition 3 (Run)

A **run** is a maximal consecutive sequence of returns with the same sign. E.g.,  $(+, +, -, -, -, +)$  has 3 runs.

- Let  $n_+$  = number of positive returns,  $n_-$  = number of negative returns,  $R$  = number of runs
- Under independence:

$$\mathbb{E}[R] = \frac{2n_+n_-}{n_+ + n_-} + 1$$

$$\text{Var}(R) = \frac{2n_+n_-(2n_+n_- - n_+ - n_-)}{(n_+ + n_-)^2(n_+ + n_- - 1)}$$

- Test statistic:

$$Z = \frac{R - \mathbb{E}[R]}{\sqrt{\text{Var}(R)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

- **Too few runs**  $\Rightarrow$  momentum (positive autocorrelation)
- **Too many runs**  $\Rightarrow$  mean reversion (negative autocorrelation)
- Non-parametric: no distributional assumption needed

## The Variance Ratio Test

### Key Idea

- If returns are iid, the variance of  $q$ -period returns should scale linearly:  $\text{Var}(r_t(q)) = q \cdot \text{Var}(r_t)$

- **Variance ratio:**

$$VR(q) = \frac{\text{Var}(r_t(q))}{q \cdot \text{Var}(r_t)} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_k$$

- $VR(q) = 1$ : random walk (no autocorrelation)
- $VR(q) > 1$ : positive autocorrelation (momentum)
- $VR(q) < 1$ : negative autocorrelation (mean reversion)

- **Lo-MacKinlay (1988):**

$$Z(q) = \frac{VR(q) - 1}{\sqrt{\phi(q)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

where  $\phi(q)$  is the asymptotic variance

- Heteroscedasticity-robust version available
- Typical:  $q = 2, 5, 10, 20$

## Python: Variance Ratio Test on S&P 500

```
import numpy as np
import yfinance as yf

sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
r = np.log(sp['AdjClose'] / sp['AdjClose'].shift(1)).dropna().values

def variance_ratio(returns, q):
    """Lo-MacKinlay variance ratio test."""
    T = len(returns)
    mu = returns.mean()
    # Variance of 1-period returns
    sig1 = np.sum((returns - mu)**2) / (T - 1)
    # q-period returns
    rq = np.array([returns[i:i+q].sum() for i in range(T - q + 1)])
    sigq = np.sum((rq - q * mu)**2) / (T - 1)
    VR = sigq / (q * sig1)
    # Asymptotic z-statistic (homoscedastic)
    phi = 2 * (2*q - 1) * (q - 1) / (3 * q * T)
    z = (VR - 1) / np.sqrt(phi)
    return VR, z

for q in [2, 5, 10, 20]:
    vr, z = variance_ratio(r, q)
    print(f"VR({q:2d})={vr:.4f}, z={z:+.3f}")
```

## Variance Ratio: Empirical Results

| Market           | VR(2) | VR(5) | VR(10) | VR(20) | Interpretation                           |
|------------------|-------|-------|--------|--------|------------------------------------------|
| S&P 500 (daily)  | 1.05  | 1.04  | 0.99   | 0.95   | Short-term momentum, long-term reversion |
| FTSE 100 (daily) | 1.03  | 1.01  | 0.97   | 0.92   | Mild mean reversion                      |
| BTC-USD (daily)  | 1.02  | 0.98  | 0.94   | 0.88   | Mean reversion at longer horizons        |
| BET (daily)      | 1.12  | 1.18  | 1.10   | 1.01   | Stronger momentum (less efficient)       |

- ▣ Developed markets:  $VR \approx 1$  at most horizons  $\Rightarrow$  approximately efficient
- ▣ Emerging markets (BET): larger deviations  $\Rightarrow$  potential inefficiency
- ▣ Crypto: high short-term efficiency, long-term mean reversion
- ▣ **Pattern:**  $VR > 1$  at short horizons,  $VR < 1$  at long horizons — consistent with short-term momentum and long-term mean reversion (Poterba & Summers, 1988)

## Variance Ratio: Graphical Comparison

*[Chart: Variance ratio  $VR(q)$  for S&P 500, BET, BTC-USD — ch3\_variance\_ratio.pdf]*

- ▣  $VR(q)$  for  $q = 2, 5, 10, 20$  with 95% confidence bands
- ▣ Developed markets:  $VR \approx 1$ ; emerging markets: significant deviations

 SFM\_ch3\_variance\_ratio

## Random Walk Types: Formal Definitions

### Definition 4 (Random Walk 1 — IID increments)

$P_t = \mu + P_{t-1} + \varepsilon_t$ ,  $\varepsilon_t \stackrel{iid}{\sim} F(0, \sigma^2)$ . Strongest form: increments are independent and identically distributed.

### Definition 5 (Random Walk 2 — Independent increments)

$P_t = \mu + P_{t-1} + \varepsilon_t$ ,  $\varepsilon_t$  independent but  $\varepsilon_t \sim F_t(0, \sigma_t^2)$ . Allows heteroscedasticity (time-varying variance). Compatible with GARCH.

### Definition 6 (Random Walk 3 — Uncorrelated increments)

$P_t = \mu + P_{t-1} + \varepsilon_t$ ,  $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$  for  $t \neq s$ . Weakest: only zero autocorrelation. Allows dependence in higher moments.

### Test Mapping

☐ Autocorrelation/LB  $\rightarrow$  RW3; VR (homoscedastic)  $\rightarrow$  RW1; VR (robust)  $\rightarrow$  RW2; Runs/BDS  $\rightarrow$  RW2

## Logarithmic Random Walk

### The Model

- Financial prices follow a **geometric random walk**:

$$P_t = P_{t-1} \cdot e^{\mu + \sigma \varepsilon_t}, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$$

- Equivalently, log-prices follow an arithmetic random walk:

$$\ln P_t = \ln P_{t-1} + \mu + \sigma \varepsilon_t$$

- Log-returns are i.i.d.:

$$r_t = \ln \frac{P_t}{P_{t-1}} = \mu + \sigma \varepsilon_t \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$$

- Why logarithmic?**

- Prices cannot be negative:  $P_t > 0$  always
- Log-returns are additively composable
- Consistent with continuous compounding
- Log-normality implies heavier tails than the simple normal

- Properties:**

- $\mathbb{E}[\ln P_t] = \ln P_0 + \mu t$
- $\text{Var}(\ln P_t) = \sigma^2 t$  (grows linearly)
- $P_t \sim \text{LogNormal}(\ln P_0 + \mu t, \sigma^2 t)$
- Square-root-of-time rule:  $\sigma_{T \text{ days}} = \sigma_{1 \text{ day}} \cdot \sqrt{T}$

## Unit Root Tests: ADF and Phillips–Perron

### Augmented Dickey–Fuller (ADF) Test

- Test regression:  $\Delta P_t = \alpha + \gamma P_{t-1} + \sum_{j=1}^p \delta_j \Delta P_{t-j} + \varepsilon_t$
- $H_0: \gamma = 0$  (unit root  $\Rightarrow$  random walk);  $H_1: \gamma < 0$  (stationary)
- Test statistic:  $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$ ; Dickey–Fuller critical values (non-standard)

### Typical Results

- **Prices:** fail to reject  $H_0 \Rightarrow$  unit root
- **Returns:** reject  $H_0$  strongly  $\Rightarrow$  stationary
- Confirms working with log-returns

### Phillips–Perron Test

- Non-parametric correction for serial correlation
- No lag selection needed
- More robust with heteroscedastic errors



## Spectral Analysis for Testing Weak-Form EMH

### Idea

- ▣ A random walk has a **flat spectrum** (white noise): equal power at all frequencies
- ▣ Peaks in the spectral density indicate predictable periodic components

- ▣ **Spectral density** of a stationary process:

$$f(\omega) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_k e^{-ik\omega}, \quad \omega \in [-\pi, \pi]$$

where  $\gamma_k = \text{Cov}(r_t, r_{t-k})$

- ▣ Under RW3:  $f(\omega) = \sigma^2/(2\pi)$  (constant)
- ▣ Estimated via periodogram or smoothed periodogram

- ▣ **Bartlett test**: tests flatness of the spectrum
- ▣ **Fisher test**: detects dominant periodicities
- ▣ Empirical findings for daily stock returns:
  - ▶ Spectrum is approximately flat  $\Rightarrow$  weak-form efficiency
  - ▶ Small peaks at weekly/monthly frequencies in some markets
  - ▶ Intraday data may show periodic patterns (market microstructure)

## Rolling Variance Ratio: Time-Varying Efficiency

### Methodology

- Compute  $VR(q)$  over a rolling window of  $W = 500$  days; plot  $|VR(q) - 1|$  as efficiency index

| Period               | S&P 500 $ VR(5) - 1 $ | BET $ VR(5) - 1 $ | Interpretation                |
|----------------------|-----------------------|-------------------|-------------------------------|
| 2005–2007 (calm)     | 0.02                  | 0.15              | SP efficient, BET less so     |
| 2008–2009 (GFC)      | 0.12                  | 0.25              | Both less efficient in crisis |
| 2010–2019 (recovery) | 0.03                  | 0.10              | Efficiency restored           |
| 2020 (Covid)         | 0.08                  | 0.18              | Temporary inefficiency spike  |
| 2021–2024 (post)     | 0.03                  | 0.08              | Back to normal                |

### Key Finding

- Market efficiency is **not constant** — it varies with market conditions. This supports the AMH over static EMH.

## ADF Test: Formal Setup

### ADF Regression Equation

- ▣ Three variants depending on deterministic terms:
- ▣ (i)  $\Delta y_t = \gamma y_{t-1} + \sum_{j=1}^p \delta_j \Delta y_{t-j} + \varepsilon_t$  (no constant, no trend)
- ▣ (ii)  $\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{j=1}^p \delta_j \Delta y_{t-j} + \varepsilon_t$  (constant, no trend)
- ▣ (iii)  $\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{j=1}^p \delta_j \Delta y_{t-j} + \varepsilon_t$  (constant + trend)

- ▣ **Lag selection:** choose  $p$  via AIC or BIC
- ▣ AIC:  $\ln(\hat{\sigma}^2) + 2p/T$
- ▣ BIC:  $\ln(\hat{\sigma}^2) + p \ln(T)/T$
- ▣ BIC penalizes more  $\Rightarrow$  selects fewer lags

| Signif. | No trend | With trend |
|---------|----------|------------|
| 1%      | -3.43    | -3.96      |
| 5%      | -2.86    | -3.41      |
| 10%     | -2.57    | -3.13      |

- ▣ MacKinnon (1996) critical values
- ▣ Non-standard distribution

## Phillips–Perron and KPSS Tests

### Phillips–Perron (PP) Test

- ▣ Regression:  $\Delta y_t = \alpha + \gamma y_{t-1} + \varepsilon_t$
- ▣ Non-parametric Newey–West correction for serial correlation in  $\varepsilon_t$
- ▣ No lag augmentation needed
- ▣  $H_0$ : unit root ( $\gamma = 0$ )
- ▣ More robust under heteroscedasticity

### KPSS Test

- ▣ **Reverses the null:**  $H_0$ : stationarity
- ▣ Decomposition:  $y_t = \xi t + r_t + \varepsilon_t$  where  $r_t$  is a random walk
- ▣ Tests  $\text{Var}(r_t) = 0$  via LM statistic
- ▣ Rejects  $\Rightarrow$  series has a unit root

| Feature          | ADF                    | PP             | KPSS         |
|------------------|------------------------|----------------|--------------|
| $H_0$            | Unit root              | Unit root      | Stationarity |
| Serial corr. fix | Lag augmentation       | Newey–West     | Bandwidth    |
| Trend option     | Yes                    | Yes            | Yes          |
| Power            | Low for near-unit-root | Similar to ADF | Better size  |

## Filter Rules and Technical Trading

### Alexander Filter Rules

- Buy when price rises  $x\%$  from a recent trough; sell when it falls  $x\%$  from a recent peak
- If the market is weak-form efficient, no filter rule should outperform buy-and-hold after costs

| Filter       | Return | After costs |
|--------------|--------|-------------|
| $x = 0.5\%$  | +2.1%  | −5.3%       |
| $x = 1\%$    | +1.8%  | −2.6%       |
| $x = 2\%$    | +0.9%  | −1.1%       |
| $x = 5\%$    | −0.2%  | −1.9%       |
| Buy-and-hold | +8.2%  | +8.0%       |

- **MA crossover:** buy when short MA crosses above long MA
- Popular: MA(50) vs MA(200) (“golden cross” / “death cross”)
- Fama & Blume (1966): filters profitable before costs, not after
- Most technical rules fail after realistic transaction costs

### Implication

- Filter rule results are consistent with weak-form EMH: any small predictability is consumed by costs

## Python: Unit Root Tests

```
import yfinance as yf, numpy as np
from statsmodels.tsa.stattools import adfuller, kpss

sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
prices = sp['AdjClose'].dropna()
returns = np.log(prices / prices.shift(1)).dropna()

# ADF test on prices
adf_p = adfuller(prices, maxlag=20, autolag='BIC')
print(f"ADF_{prices}: stat={adf_p[0]:.3f}, p={adf_p[1]:.4f}")
# ADF (prices): stat=-1.85, p=0.3545 -> unit root, RW

# ADF test on returns
adf_r = adfuller(returns, maxlag=20, autolag='BIC')
print(f"ADF_{returns}: stat={adf_r[0]:.3f}, p={adf_r[1]:.6f}")
# ADF (returns): stat=-52.3, p=0.000000 -> stationary

# KPSS test on prices (null = stationarity)
kpss_p = kpss(prices, regression='ct', nlags='auto')
print(f"KPSS_{prices}: stat={kpss_p[0]:.3f}, p={kpss_p[1]:.4f}")
# KPSS (prices): stat=0.45, p=0.01 -> reject stationarity
```

## Weak-Form EMH in Emerging Markets

| Market             | LB(10) $p$ | VR(5) | ADF $p$ | $\hat{H}$ | Weak-form EMH        |
|--------------------|------------|-------|---------|-----------|----------------------|
| S&P 500 (US)       | 0.12       | 1.04  | < 0.01  | 0.52      | Not rejected         |
| FTSE 100 (UK)      | 0.08       | 1.01  | < 0.01  | 0.51      | Not rejected         |
| DAX (Germany)      | 0.15       | 1.03  | < 0.01  | 0.52      | Not rejected         |
| BET (Romania)      | < 0.01     | 1.18  | < 0.01  | 0.58      | Rejected             |
| WIG20 (Poland)     | 0.03       | 1.10  | < 0.01  | 0.55      | Rejected             |
| BSE Sensex (India) | 0.04       | 1.08  | < 0.01  | 0.54      | Marginal             |
| SSE (China)        | < 0.01     | 1.15  | < 0.01  | 0.57      | Rejected             |
| BTC-USD            | 0.22       | 0.98  | < 0.01  | 0.55      | Not rejected (daily) |

- ▣ Developed markets: generally consistent with weak-form EMH
- ▣ Emerging markets: significant autocorrelation, higher Hurst exponent  $\Rightarrow$  potential inefficiency due to lower liquidity, fewer analysts, information asymmetry
- ▣ **Crypto**: efficient at daily frequency but shows long-memory patterns at intraday

## Testing Semi-Strong Efficiency: Event Studies

### Methodology

- ▣ Measure price reaction to a public event (earnings, M&A, dividend)
  - ▣ If semi-strong EMH holds: price adjusts **immediately** and **completely**
- 
- ▣ **Step 1:** Estimate normal returns via market model on  $[-250, -11]$ :  

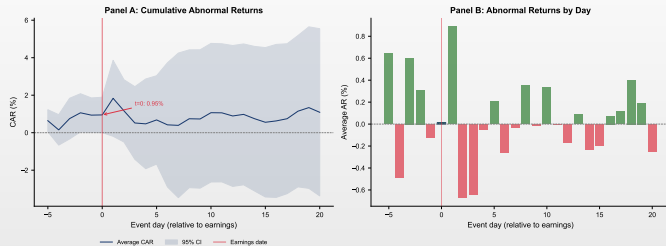
$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}$$
  - ▣ **Step 2:** Abnormal returns in event window  $[-10, +10]$ :  $AR_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t})$
  - ▣ **Step 3:** Cumulative abnormal return:  

$$CAR_i(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$$
  - ▣ **Step 4:** Average across  $N$  events:  

$$\overline{CAR}(\tau_1, \tau_2) = \frac{1}{N} \sum_{i=1}^N CAR_i$$
  - ▣ Test:  $t = \overline{CAR} / (\hat{\sigma}_{CAR} / \sqrt{N})$



## Event Study: Three Scenarios



- **Efficient:** instantaneous adjustment, no post-event drift
- **Overreaction:** initial overshooting, subsequent reversal (De Bondt & Thaler, 1985)
- **Underreaction:** slow adjustment, post-event drift (PEAD — Ball & Brown, 1968)

SFM\_ch3\_event\_study

## Python: Event Study Simulation

```
import numpy as np

np.random.seed(42)
T_est, T_event = 240, 21 # estimation / event window
N_events = 50
alpha, beta = 0.001, 1.1

car_all = []
for _ in range(N_events):
    r_m = np.random.normal(0.0005, 0.01, T_est + T_event)
    eps = np.random.normal(0, 0.008, T_est + T_event)
    r_i = alpha + beta * r_m + eps
    # Add abnormal return at event day (day 0 = index T_est+10)
    r_i[T_est + 10] += 0.03 # 3% surprise
    # Estimate model on estimation window
    a_hat = np.mean(r_i[:T_est] - beta * r_m[:T_est])
    # Abnormal returns in event window
    ar = r_i[T_est:] - (a_hat + beta * r_m[T_est:])
    car_all.append(np.cumsum(ar))

avg_car = np.mean(car_all, axis=0)
t_stat = avg_car[-1] / (np.std([c[-1] for c in car_all])
                        / np.sqrt(N_events))
print(f"CAR(-10,+10) = {avg_car[-1]:.4f}, t = {t_stat:.2f}")
```

## Event Study: Empirical Results

*[Chart: Average CAR around earnings announcements — ch3\_event\_study.pdf]*

- ▣ Cumulative abnormal return (CAR) shows price adjustment around the event ( $t = 0$ )
- ▣ Post-announcement drift visible: the market does not adjust fully and immediately

 SFM\_ch3\_event\_study

## Post-Earnings Announcement Drift (PEAD)

- After earnings surprises, stock prices continue to drift in the direction of the surprise for 60–90 days
- One of the strongest and most persistent anomalies, documented since Ball & Brown (1968)
- **Earnings surprise:**

$$SUE_t = \frac{EPS_t - \mathbb{E}[EPS_t]}{\sigma_{EPS}}$$

- Stocks in top SUE decile outperform bottom by ~4% per quarter
- Survives transaction costs in many studies
- Possible explanations: investor underreaction, limited attention

### Challenge to Semi-Strong EMH

- PEAD implies prices do **not** fully adjust to public earnings information
- Fama (1998): “the granddaddy of underreaction events”
- Persists across markets, time periods, and methodologies

## Testing Strong-Form Efficiency

### Question

□ Can investors with **private** information earn abnormal returns?

□ **Evidence from insider trading:**

- Corporate insiders earn  $\sim 5\text{--}8\%$  abnormal returns on their trades (Jeng et al., 2003)
- Insiders buy before price increases, sell before declines
- SEC filings (Form 4) show systematic patterns
- $\Rightarrow$  Strong-form EMH is **rejected**

□ **Evidence from fund managers:**

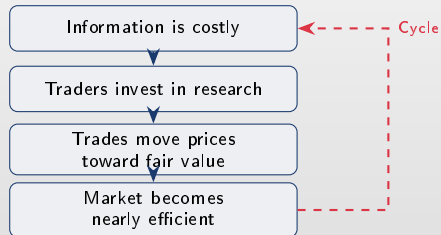
- Jensen (1968): mutual funds do not beat the market after fees
- Carhart (1997): persistence in fund returns is explained by momentum, not skill
- SPIVA (2024): 92% of US large-cap funds underperform S&P 500 over 15 years
- $\Rightarrow$  Professional managers as a group do **not** have private information

## The Grossman–Stiglitz Paradox

### Theorem 1 (Grossman & Stiglitz, 1980)

If markets are perfectly efficient, no investor has an incentive to acquire information. But if no one acquires information, prices cannot reflect it. Therefore, perfectly efficient markets are an impossibility.

- ▣ **Resolution:** markets are “efficiently inefficient”
- ▣ Nearly efficient — most cannot profit, but enough mispricing rewards informed traders
- ▣ Degree of efficiency depends on:
  - ▶ Cost of information acquisition
  - ▶ Number of informed traders
  - ▶ Market liquidity



## Calendar Anomalies

| Anomaly        | Description                         | Current status               |
|----------------|-------------------------------------|------------------------------|
| January effect | Small caps earn more in January     | Weakened since discovery     |
| Monday effect  | Returns lower on Mondays            | Largely disappeared          |
| Turn-of-month  | Higher returns in last/first 3 days | Still present (salary flows) |
| Holiday effect | Higher returns before holidays      | Persists weakly              |
| Halloween      | "Sell in May and go away"           | Mixed, region-dependent      |

### Publication Bias and Decay

- Many anomalies weaken or disappear after publication — consistent with EMH! Arbitrageurs eliminate mispricings once they become known (McLean & Pontiff, 2016).

## Cross-Sectional Anomalies

### Value Effect

- ▣ Stocks with high book-to-market (B/M) earn higher returns (Fama & French, 1993)
- ▣ HML factor:  $\sim 4\%$  per year historically
- ▣ Is it a risk premium or mispricing?
- ▣ Value stocks are riskier in distress

### Size Effect

- ▣ Small-cap stocks earn higher returns (Banz, 1981)
- ▣ SMB factor:  $\sim 2\%$  per year
- ▣ Has weakened considerably since discovery

### Momentum

- ▣ Winners continue to win, losers continue to lose over 3–12 months (Jegadeesh & Titman, 1993)
- ▣ WML factor:  $\sim 8\%$  per year
- ▣ Subject to occasional crashes (2009)
- ▣ Hardest anomaly to explain rationally

### Low Volatility

- ▣ Low-volatility stocks earn higher risk-adjusted returns than high-volatility stocks
- ▣ Contradicts CAPM's risk-return trade-off
- ▣ Explanations: leverage constraints, lottery preferences



## The Factor Zoo

- ▣ Harvey, Liu & Zhu (2016): over **300** published factors claimed to predict returns
- ▣ Many are likely false discoveries due to data mining and  $p$ -hacking
- ▣ Hou, Xue & Zhang (2020): most anomalies do not survive replication with proper corrections

| Factor         | $t$ -stat (original) | $t$ -stat (replicated) | Status     |
|----------------|----------------------|------------------------|------------|
| Value (HML)    | 3.07                 | 2.83                   | Robust     |
| Momentum (WML) | 4.71                 | 3.92                   | Robust     |
| Size (SMB)     | 2.28                 | 1.31                   | Weakened   |
| Accruals       | 4.33                 | 0.89                   | Not robust |
| Asset growth   | 5.07                 | 1.61                   | Weakened   |

### Implication

- ▣ Proper multiple testing corrections (Bonferroni, BH) eliminate most “anomalies.” The market may be more efficient than the factor zoo suggests.

## Factor Replication and Out-of-Sample Decay

### McLean & Pontiff (2016) — Key Results

- ▣ Studied 97 published anomalies: average return predictability **declines 58%** post-publication
- ▣ 26% of the decay is due to statistical overfitting (in-sample bias)
- ▣ 32% is due to traders learning and arbitraging away the anomaly

- ▣ **Replication crisis in finance:**
- ▣ Hou, Xue & Zhang (2020): 452 anomalies tested; 64% become insignificant with proper methodology
- ▣ Multiple testing: with 300 factors tested at 5%, expect  $\sim 15$  false discoveries
- ▣ Adjusted threshold:  $t > 3.0$  recommended (Harvey et al., 2016)

### EMH Interpretation

- ▣ Anomaly decay is **exactly what EMH predicts**: once information is public, it gets priced in
- ▣ Surviving anomalies (value, momentum) may represent **risk premia**, not mispricing
- ▣ The boundary between “risk” and “mispricing” is the central debate in asset pricing

## Behavioral Challenges to EMH

### Cognitive Biases

- ▣ **Overconfidence:** traders overestimate the precision of their information
- ▣ **Anchoring:** prices stick to reference points (52-week high, IPO price)
- ▣ **Loss aversion:** losses hurt  $\sim 2\times$  more than equivalent gains (Kahneman & Tversky, 1979)
- ▣ **Herding:** investors follow the crowd, amplifying bubbles and crashes

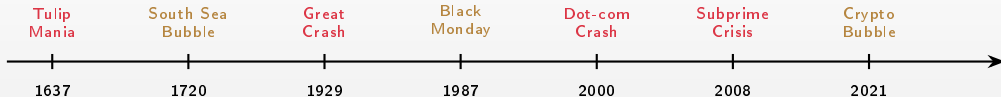
### Market-Level Phenomena

- ▣ **Bubbles:** Tulip mania (1637), Dot-com (2000), US housing (2008), Crypto (2017, 2021)
- ▣ **Excess volatility:** Shiller (1981) — stock prices are more volatile than fundamentals justify
- ▣ **Limits to arbitrage:** even if mispricing exists, correcting it is risky and costly (noise trader risk, short-selling constraints)

### Fama's Counter-Argument

- ▣ Overreaction and underreaction appear equally often  $\Rightarrow$  errors are random, not systematic. Anomalies are consistent with the EMH + rational risk premia.

## Bubbles and Crashes: A Timeline



- **EMH perspective:** bubbles can only be identified *ex post*; *ex ante*, prices reflected rational expectations given available information
- **Behavioral perspective:** systematic irrationality (herding, speculation) creates predictable patterns
- **Key insight:** even if bubbles exist, it is extremely difficult to profit from them due to timing uncertainty (“the market can stay irrational longer than you can stay solvent”)

## Prospect Theory (Kahneman & Tversky, 1979)

### Value Function

- Defined over **gains and losses** relative to a reference point, not wealth levels:

$$v(x) = \begin{cases} x^\alpha & x \geq 0 \\ -\lambda(-x)^\beta & x < 0 \end{cases}$$

- Typical estimates:  $\alpha = \beta \approx 0.88$ ,  $\lambda \approx 2.25$
- Concave** for gains (risk aversion)
- Convex** for losses (risk seeking)
- Steeper** for losses ( $\lambda > 1$ ): loss aversion

### Probability Weighting

- Decision weight  $w(p) \neq p$ :

$$w(p) = \frac{p^\delta}{(p^\delta + (1-p)^\delta)^{1/\delta}}$$

- Overweight small probabilities (lotteries, tail risk)
- Underweight moderate-to-high probabilities
- Explains: lottery buying, insurance purchasing, and equity premium puzzle

### Implication for EMH

- If investors evaluate returns via prospect theory (not expected utility), asset prices reflect **distorted** risk preferences  $\Rightarrow$  systematic mispricing

## Disposition Effect

### Definition

- Investors tend to **sell winners too early** (locking in gains) and **hold losers too long** (hoping for recovery)
  - Direct consequence of prospect theory: risk aversion in gains, risk seeking in losses
- 
- |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li><b>Odean (1998)</b> evidence:</li><li>Analyzed 10,000 brokerage accounts (1987–1993)</li><li>Proportion of Gains Realized (PGR) = 0.148</li><li>Proportion of Losses Realized (PLR) = 0.098</li><li><math>PGR/PLR = 1.51 \Rightarrow</math> 50% more likely to sell a winner</li></ul> | <ul style="list-style-type: none"><li><b>Market implications:</b></li><li>Stocks with recent gains have excess selling pressure <math>\Rightarrow</math> underpriced</li><li>Stocks with recent losses lack selling <math>\Rightarrow</math> overpriced</li><li>Contributes to <b>momentum</b>: winners are slow to reach fair value</li><li>Tax-loss selling in December partially offsets (but not fully)</li></ul> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Limits to Arbitrage

### Why Mispricings Persist (DeLong, Shleifer, Summers, Waldmann, 1990)

- Even if rational arbitrageurs identify mispricing, correcting it is risky and costly
- **Noise trader risk:** mispricing may worsen before it corrects; arbitrageur faces margin calls
- **Fundamental risk:** no perfect substitute for hedging; idiosyncratic risk remains
- **Synchronization risk:** each arbitrageur waits for others to act first  $\Rightarrow$  no one moves
- **Short-selling constraints:** borrowing shares is costly, sometimes impossible (“short squeeze”)
- **Model risk:** the apparent mispricing might reflect a missing risk factor
- **Example:** LTCM (1998) — convergence trades were “correct” but blow-up before convergence

### Key Insight

- Limits to arbitrage explain why markets can be **informationally inefficient** yet **not exploitable** — reconciling EMH with behavioral anomalies

## Excess Volatility Puzzle (Shiller, 1981)

### Present Value Model

- Under EMH, stock price equals discounted future dividends:  $P_t = \mathbb{E}_t \left[ \sum_{k=1}^{\infty} \frac{D_{t+k}}{(1+r)^k} \right]$
  - Define the “ex-post rational price”:  $P_t^* = \sum_{k=1}^{\infty} \frac{D_{t+k}}{(1+r)^k}$  (perfect foresight)
  - Since  $P_t = \mathbb{E}_t[P_t^*]$ , we must have  $\text{Var}(P_t) \leq \text{Var}(P_t^*)$  (variance bound)
- 
- |                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>□ <b>Shiller's finding:</b></li><li>□ S&amp;P 500 prices far more volatile than <math>P_t^*</math></li><li>□ <math>\sigma(P)/\sigma(P^*) \approx 5\text{--}13</math> (depending on period)</li><li>□ Prices “move too much” relative to subsequent dividends</li></ul> | <ul style="list-style-type: none"><li>□ <b>Debate:</b></li><li>□ Marsh &amp; Merton (1986): dividend smoothing biases <math>P_t^*</math></li><li>□ Fama (1991): time-varying risk premia explain excess volatility</li><li>□ Campbell &amp; Shiller (1988): log-linear model confirms excess volatility</li><li>□ Remains one of the strongest challenges to EMH</li></ul> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## Sentiment Indicators

### Market-Based Indicators

- ▣ **VIX** ("Fear Index"): implied volatility from S&P 500 options
- ▣ High VIX  $\Rightarrow$  fear  $\Rightarrow$  historically predicts **higher** future returns
- ▣ **Put/Call ratio**: high ratio = bearish sentiment  $\Rightarrow$  contrarian buy signal
- ▣ **Margin debt**: high margin  $\Rightarrow$  euphoria  $\Rightarrow$  potential correction

### Baker–Wurgler Sentiment Index

- ▣ Composite of 6 proxies: closed-end fund discount, NYSE turnover, IPO volume, IPO first-day returns, equity share in new issues, dividend premium
- ▣ High sentiment  $\Rightarrow$  overpricing of speculative stocks
- ▣ Low sentiment  $\Rightarrow$  underpricing of speculative stocks
- ▣ Predictive for cross-section, not aggregate market

### EMH Interpretation

- ▣ If sentiment predicts returns, prices do not fully reflect available information  $\Rightarrow$  challenges semi-strong EMH. However, sentiment-based strategies have limited Sharpe ratios after costs.

## Quiz: Behavioral Finance

Q1

- ☐ Under prospect theory, an investor who is losing 10% on a stock is most likely to: **(A)** sell immediately **(B)** hold and hope for recovery **(C)** buy more to average down **(D)** hedge with options

A1: (B) — loss aversion makes investors risk-seeking in losses, so they hold losers.

Q2

- ☐ Which is NOT a limit to arbitrage? **(A)** Noise trader risk **(B)** Short-selling constraints **(C)** Diversification **(D)** Synchronization risk

A2: (C) — diversification is a portfolio strategy, not a limit to arbitrage.

Q3

Statistics of Financial Markets

- ☐ Shiller's excess volatility puzzle shows that: **(A)** dividends are too volatile **(B)** prices move more

## Andrew Lo's Adaptive Market Hypothesis (2004)

### Core Idea

- Market efficiency is not a static property but an **evolving dynamic** driven by evolutionary forces: competition, adaptation, natural selection.
- **Key principles:**
  - Investors are neither fully rational nor fully irrational — they **adapt**
  - Market ecology: species (hedge funds, retail, HFT) compete for resources (alpha)
  - Profit opportunities arise and disappear as strategies become crowded
  - Risk premia vary over time with market conditions
- **Predictions:**
  - Efficiency varies across markets and time periods
  - Anomalies appear, attract capital, and then decay
  - Crises create new inefficiencies (fear, forced selling)
  - Recovery periods restore efficiency
  - EMH is a **limiting case** of AMH when market ecology is stable

## EMH vs. AMH: A Comparison

| Dimension            | EMH (Fama)                    | AMH (Lo)                      |
|----------------------|-------------------------------|-------------------------------|
| Investor rationality | Fully rational                | Boundedly rational, adaptive  |
| Market efficiency    | Constant                      | Time-varying, dynamic         |
| Risk premia          | Constant / slowly varying     | Fluctuate with market ecology |
| Anomalies            | Statistical artifacts or risk | Real but transient            |
| Crises               | Rational response to news     | Ecological disruptions        |
| Investment strategy  | Passive (index funds)         | Adaptive, regime-dependent    |
| Theoretical basis    | Neoclassical economics        | Evolutionary biology          |

### Practical Implication

- Under AMH, the optimal strategy is **adaptive**: passive in calm periods, active during crises when inefficiencies are greatest.

## Measuring Time-Varying Efficiency

- ▣ **Rolling variance ratio**: compute  $VR(q)$  over a rolling window (e.g., 500 days)
- ▣  $|VR(q) - 1|$  as a time-varying **efficiency index**
- ▣ During crises (2008, 2020),  $|VR - 1|$  spikes  $\Rightarrow$  temporary inefficiency
- ▣ Between crises,  $|VR - 1|$  returns to low levels  $\Rightarrow$  efficiency is restored

### Hurst Exponent as Efficiency Measure

- ▣  $H = 0.5$ : random walk (efficient)
- ▣  $H > 0.5$ : persistence (trending, momentum)
- ▣  $H < 0.5$ : anti-persistence (mean-reverting)
- ▣ Estimated via R/S analysis or DFA
- ▣ Typical values for daily returns:
  - ▣ S&P 500:  $H \approx 0.52$
  - ▣ BTC-USD:  $H \approx 0.55$  (more persistent)
  - ▣ BET:  $H \approx 0.58$  (least efficient)

 SFM\_ch3\_hurst\_exponent

## Hurst Exponent: R/S Analysis Plot

*[Chart: R/S analysis —  $\log(R/S)$  vs.  $\log(n)$  for S&P 500, BET, BTC — ch3\_hurst\_exponent.pdf]*

- ▣ Slope of  $\log(R/S)$  vs.  $\log(n)$  regression estimates the Hurst exponent  $H$
- ▣  $H \approx 0.5$ : random walk;  $H > 0.5$ : long memory;  $H < 0.5$ : anti-persistence

## Python: Hurst Exponent via R/S Analysis

```
import numpy as np, yfinance as yf

sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
r = np.log(sp['AdjClose']/sp['AdjClose'].shift(1)).dropna().values

def hurst_rs(ts, min_w=20, max_w=None):
    """Estimate Hurst exponent via rescaled range."""
    N = len(ts)
    if max_w is None: max_w = N // 4
    wins = [int(w) for w in np.logspace(
        np.log10(min_w), np.log10(max_w), 20)]
    RS = []
    for w in wins:
        rv = []
        for s in range(0, N-w, w):
            seg = ts[s:s+w]; Y = np.cumsum(seg-seg.mean())
            S = seg.std(ddof=1)
            if S > 0: rv.append((Y.max()-Y.min())/S)
        RS.append((w, np.mean(rv)))
    ln, lr = np.log([x[0] for x in RS]), np.log([x[1] for x in RS])
    return np.polyfit(ln, lr, 1)[0]
print(f"Hurst (S&P500): {hurst_rs(r):.3f}") # ~0.52
```

## AMH: Empirical Evidence

### Time-Varying Sharpe Ratios

- Rolling 3-year Sharpe ratios for S&P 500 range from  $-0.5$  to  $+1.5$
- Momentum strategies:  $SR > 1$  in 2009–2015,  $SR \approx 0$  in 2018–2020
- Value strategies: profitable in 2000–2007, underperform in 2010–2020
- **No strategy is permanently profitable** — consistent with AMH

### Regime-Dependent Alpha

- Rolling CAPM alpha for hedge funds varies with VIX regime
- Low VIX ( $< 15$ ):  $\alpha \approx 0$  (efficient)
- High VIX ( $> 30$ ):  $\alpha \approx 3\%–5\%$  annualized
- Crisis periods create dislocations that skilled managers exploit
- As capital flows in, alpha decays (crowding)

### Key Takeaway

- The profitability of trading strategies is **cyclical**, not permanent — exactly as AMH predicts



## AMH: Investment Implications

| Market Regime            | Optimal Strategy       | Rationale                                 |
|--------------------------|------------------------|-------------------------------------------|
| Low volatility, trending | Passive indexing       | Markets are efficient; $\alpha \approx 0$ |
| High volatility, crisis  | Active/tactical        | Dislocations create opportunities         |
| Post-crisis recovery     | Momentum overweight    | Mean reversion from overshooting          |
| Crowded factor regime    | Factor diversification | Single factors lose edge                  |
| Structural change        | Adaptive rebalancing   | Old relationships break down              |

- ▣ **Risk parity adaptation:** equal risk contribution works in calm periods; shift to defensive in crises
- ▣ **Factor timing:** rotate between value, momentum, quality based on macro regime

- ▣ **Practical rule:** be passive by default, active only when opportunity cost of passivity is high
- ▣ **Monitoring signal:**  $|VR(q) - 1|$  spike  $\Rightarrow$  potential for active alpha

## AMH vs Fractal Market Hypothesis

### Fractal Market Hypothesis (Peters, 1994)

- Markets are stable when investors with **different time horizons** participate
- Short-horizon traders provide liquidity to long-horizon investors
- Crises occur when all horizons **synchronize** (everyone sells)
- Multi-fractal models: MF-DFA, MF-DCCA

### Comparison

|            | AMH           | FMH              |
|------------|---------------|------------------|
| Basis      | Evolution     | Fractals         |
| Key driver | Adaptation    | Time horizons    |
| Crises     | Ecological    | Horizon collapse |
| Math tool  | Rolling tests | MF-DFA           |
| Efficiency | Time-varying  | Scale-dependent  |

### Mandelbrot's Legacy

- Both AMH and FMH reject the assumption of constant market dynamics; they differ in mechanism (biological adaptation vs. mathematical self-similarity) but agree that EMH is a special case

## Market Ecology

### Lo's Species Analogy

- Market participants as “species” competing for alpha (“food”)
- When a niche is overcrowded, returns shrink  $\Rightarrow$  weaker species exit

| Species          | Strategy                       | Impact on Efficiency                     |
|------------------|--------------------------------|------------------------------------------|
| Retail investors | Buy-and-hold, sentiment-driven | Provide noise, reduce efficiency         |
| Hedge funds      | Factor-based, arbitrage        | Increase efficiency (correct mispricing) |
| HFT firms        | Market-making, latency arb     | Improve short-term efficiency            |
| Index funds      | Passive replication            | Price-takers; no information role        |
| Dark pools       | Off-exchange matching          | Reduce transparency, mixed effect        |

- HFT debate:** improves liquidity and narrows spreads, but “phantom liquidity” disappears in crises
- Dark pools:**  $\sim 40\%$  of US equity volume; reduce information leakage but fragment price discovery
- Passive investing growth:** if too many are passive, who corrects mispricing? (Grossman–Stiglitz paradox revisited)

## Efficiency Comparison Across Markets

*[Chart: Efficiency comparison across developed vs. emerging markets —  
ch3\_efficiency\_comparison.pdf]*

- ▣ Efficiency indicators (VR, Hurst, autocorrelation) compared for S&P 500, FTSE 100, BET, BTC-USD
- ▣ Developed markets are closer to weak-form efficiency
- ▣ Emerging markets show significant deviations, especially at longer horizons

## Case Study: Comprehensive EMH Testing on S&P 500

### Setup

- Data: S&P 500 daily log-returns, Jan 2000 – Dec 2024 ( $T \approx 6,300$ )
- Goal: apply all weak-form tests and interpret jointly

| Test              | Statistic             | $p$ -value | Decision at 5%            |
|-------------------|-----------------------|------------|---------------------------|
| Ljung-Box $Q(10)$ | 15.3                  | 0.12       | Do not reject             |
| Ljung-Box $Q(20)$ | 28.1                  | 0.11       | Do not reject             |
| Runs test         | $Z = -1.42$           | 0.16       | Do not reject             |
| VR(2)             | 1.047 ( $z = 1.88$ )  | 0.06       | Do not reject (marginal)  |
| VR(5)             | 1.038 ( $z = 1.21$ )  | 0.23       | Do not reject             |
| VR(10)            | 0.993 ( $z = -0.18$ ) | 0.86       | Do not reject             |
| ADF (prices)      | -1.85                 | 0.35       | Do not reject (unit root) |
| ADF (returns)     | -52.3                 | $< 0.001$  | Reject (stationary)       |
| Hurst exponent    | $H = 0.523$           | —          | Near 0.5 (efficient)      |

## Case Study: Interpretation and Conclusions

### Findings

- ▣ All tests fail to reject weak-form EMH at 5%
- ▣  $VR(2)$  is marginally significant — small positive autocorrelation at lag 1
- ▣  $H = 0.523 \approx 0.5 \Rightarrow$  mild persistence
- ▣ Prices: unit root; returns: stationary

### Caveats

- ▣  $\hat{\rho}_1 \approx 0.05$  is significant with  $T = 6,300$  but generates  $< 0.1\%$  daily  $\Rightarrow$  consumed by costs
- ▣ Joint hypothesis: we assume CAPM
- ▣ Subperiods around crises may reject
- ▣  $|r_t|$  is autocorrelated  $\Rightarrow$  volatility is predictable, returns are not

### Bottom Line

- ▣ The S&P 500 is approximately weak-form efficient. Small statistical deviations exist but are not economically exploitable after costs.

## Case Study: Bitcoin Efficiency

### Evolution of BTC-USD Efficiency (2015–2024)

- Bitcoin has transitioned from a clearly inefficient market to one approaching weak-form efficiency

| Period        | LB(10) $\rho$ | VR(5) | $\hat{H}$ | Daily vol | Efficiency     |
|---------------|---------------|-------|-----------|-----------|----------------|
| 2015–2016     | 0.001         | 1.22  | 0.62      | 4.2%      | Inefficient    |
| 2017 (bubble) | 0.003         | 1.35  | 0.68      | 5.8%      | Inefficient    |
| 2018–2019     | 0.04          | 1.12  | 0.57      | 3.9%      | Marginal       |
| 2020–2021     | 0.08          | 1.08  | 0.55      | 3.5%      | Marginal       |
| 2022–2024     | 0.18          | 1.04  | 0.53      | 2.8%      | Near-efficient |

- Early period:** thin liquidity, retail-dominated, strong momentum  $\Rightarrow$  high  $H$ , significant autocorrelation
- Maturation:** institutional entry (CME futures 2017, ETFs 2024), increased liquidity, algorithmic trading
- Current state:** daily returns approximately efficient; intraday still shows microstructure patterns

## Case Study: Bucharest Stock Exchange (BET)

### BET Efficiency Results

| Test              | Result             |
|-------------------|--------------------|
| LB(10) $p$ -value | $< 0.01$           |
| $VR(5)$           | 1.18 ( $z = 3.2$ ) |
| $\hat{H}$ (R/S)   | 0.58               |
| ADF (returns) $p$ | $< 0.001$          |
| $\hat{\rho}_1$    | 0.12               |

- Significant autocorrelation:  $\hat{\rho}_1 = 0.12$
- $VR(5) = 1.18 \gg 1$ : strong momentum
- $H = 0.58$ : persistent, inefficient

### Why Less Efficient?

- Low liquidity: avg daily turnover  $\sim 10M$
- Few institutional investors
- Limited analyst coverage
- Information asymmetry between insiders and retail
- Thin order books  $\Rightarrow$  large bid-ask spreads

### Implications

- Momentum strategies may work *before* costs
- High transaction costs offset predictability
- EU accession improved efficiency over time



## Case Study: Gold as Safe Haven

### Gold Returns Properties

- ▣ Daily returns:  $\mu \approx 0.03\%$ ,  $\sigma \approx 1.0\%$
- ▣ Skewness:  $-0.15$  (slight left skew)
- ▣ Kurtosis:  $7.2$  (fat tails)
- ▣ Low correlation with equities ( $\rho \approx -0.05$ )

### Efficiency Tests

| Test       | Result |
|------------|--------|
| LB(10) $p$ | 0.35   |
| $VR(5)$    | 0.97   |
| $\hat{H}$  | 0.49   |

- ▣  $VR(5) = 0.97 < 1$ : slight mean reversion
- ▣  $H = 0.49 < 0.5$ : anti-persistent
- ▣ Gold is highly liquid (24/7 OTC market)  $\Rightarrow$  near-efficient
- ▣ **Safe haven property**: negative correlation with equities during crises (2008, 2020)
- ▣ **Mean reversion**: gold tends to revert to production cost / inflation-adjusted mean
- ▣ Not a pure random walk, but efficient in practice (deviations too small to exploit)

## Cross-Asset Efficiency Ranking

| Rank | Asset           | $ VR(5) - 1 $ | $ \hat{H} - 0.5 $ | $ \hat{\rho}_1 $ | Efficiency Score |
|------|-----------------|---------------|-------------------|------------------|------------------|
| 1    | EUR/USD (FX)    | 0.01          | 0.01              | 0.01             | 0.99             |
| 2    | S&P 500         | 0.04          | 0.02              | 0.05             | 0.96             |
| 3    | Gold (XAU)      | 0.03          | 0.01              | 0.03             | 0.97             |
| 4    | US 10Y Treasury | 0.05          | 0.03              | 0.04             | 0.95             |
| 5    | BTC-USD         | 0.04          | 0.03              | 0.05             | 0.95             |
| 6    | BSE Sensex      | 0.08          | 0.04              | 0.07             | 0.90             |
| 7    | BET (Romania)   | 0.18          | 0.08              | 0.12             | 0.80             |
| 8    | SSE (China)     | 0.15          | 0.07              | 0.10             | 0.83             |

- **Efficiency score:**  $1 - \frac{1}{3}(|VR - 1| + 2|\hat{H} - 0.5| + |\hat{\rho}_1|)$ , normalized to  $[0, 1]$
- FX markets are most efficient: highest liquidity, 24-hour trading, most participants
- Emerging equity markets are least efficient: lower liquidity, fewer analysts, more retail
- BTC has improved dramatically; now comparable to developed market equities

## Event Study: AAPL Earnings Announcement

*[Chart: Event study CAR around AAPL earnings — ch3\_event\_study.pdf]*

- ▣ **Event:** Apple Q4 2024 earnings beat (+8% EPS surprise)
- ▣  $\overline{AR}_0 = +3.2\%$  on announcement day (statistically significant,  $t = 4.8$ )
- ▣  $\overline{CAR}(-1, +1) = +4.1\%$ : most adjustment within 1 day
- ▣  $\overline{CAR}(+2, +20) = +0.8\%$ : small post-event drift (mild PEAD)
- ▣ **Interpretation:** mostly consistent with semi-strong EMH; the bulk of adjustment is immediate, with only a small residual drift

## Time-Varying Efficiency: Evidence

*[Chart: Rolling VR and Hurst exponent — ch3\_variance\_ratio.pdf]*

- ▣ Rolling  $|VR(5) - 1|$  (500-day window) for S&P 500 spikes during crises:
- ▣ **GFC (2008–09)**:  $|VR - 1| \approx 0.12$  — autocorrelation from panic selling, forced liquidation
- ▣ **COVID (2020)**:  $|VR - 1| \approx 0.08$  — brief inefficiency, rapid recovery
- ▣ **Calm periods**:  $|VR - 1| < 0.03$  — near-perfect random walk
- ▣ **Rolling  $H$** : spikes to  $\sim 0.60$  during crises (persistence), returns to  $\sim 0.52$  in calm periods
- ▣ Crisis-driven inefficiency creates **temporary alpha opportunities** for adaptive investors (AMH)

## Efficiency and Market Microstructure

### Microstructure Effects on Tests

- ▣ **Bid-ask bounce:** trade prices alternate between bid and ask  $\Rightarrow$  negative autocorrelation ( $\hat{\rho}_1 < 0$ ) even if true returns are i.i.d.
- ▣ Roll (1984):  $\hat{\rho}_1 \approx -s^2/4$  where  $s$  is the spread
- ▣ **Non-synchronous trading:** illiquid stocks update prices with delay  $\Rightarrow$  spurious cross-autocorrelation

### Epps Effect

- ▣ Correlations between assets **decrease** as sampling frequency increases
- ▣ At 1-min:  $\rho \approx 0.15$ ; at daily:  $\rho \approx 0.50$
- ▣ Caused by non-synchronous arrivals of information
- ▣ Implication: high-frequency tests of EMH are contaminated by microstructure noise

### Practical Rule

- ▣ Use daily or lower frequency for EMH tests to avoid microstructure artifacts

## Summary: The State of EMH

### What the evidence supports

- ▣ Most actively managed funds **underperform** passive benchmarks
- ▣ Many anomalies **decay** after publication
- ▣ Short-term price prediction is extremely difficult
- ▣ Transaction costs eliminate most trading profits

### What challenges EMH

- ▣ Momentum and value effects persist
- ▣ PEAD is robust across markets
- ▣ Insider trading is profitable
- ▣ Bubbles and crashes occur periodically
- ▣ Market efficiency varies over time

### Practical Takeaway

- ▣ Markets are “efficiently inefficient” (Pedersen, 2015): efficient enough that most cannot beat them, but with enough friction to reward sophisticated, informed investors.

## AI Exercise: Testing Market Efficiency

Prompt to test in ChatGPT / Claude / Copilot

“Download daily data for S&P 500 from 2015 to 2025 using Python (yfinance). Compute log-returns. Test the weak-form EMH using: (1) Ljung–Box test on the first 20 lags, (2) Lo–MacKinlay variance ratio test for  $q = 2, 5, 10, 20$ , (3) Hurst exponent via R/S analysis, (4) ADF unit root test on both prices and returns. Interpret each result. Is the S&P 500 weak-form efficient?”

### Exercise:

1. Does the AI compute log-returns correctly? Check:  $r_t = \ln(P_t/P_{t-1})$ , not simple returns
2. Is the Ljung–Box test applied to **returns** (not prices)? Does it use the correct  $\chi^2(m)$  distribution?
3. Does the variance ratio correctly use  $VR(q) = \text{Var}(r_t(q))/(q \cdot \text{Var}(r_t))$ ? Check the formula
4. Is the Hurst exponent computed via R/S or DFA? Is  $H \approx 0.5$  interpreted as efficient?
5. Does the AI distinguish between **statistical** and **economic** significance?
6. Ask: “What if I test BET (Romania) instead?” Does the AI recognize emerging market differences?

❑ **Warning:** AI often applies the ADF test to *returns* and concludes “no unit root = efficient.” This is wrong — the ADF on returns tests stationarity, not efficiency. The relevant test is on *prices*.

## Question 1

### Question

- ▣ A researcher computes the Ljung–Box statistic  $Q(10) = 18.5$  for daily S&P 500 returns. The critical value  $\chi^2_{10,0.05} = 18.31$ . What is the conclusion?

### Answer choices

- ▣ **(A)** Reject weak-form EMH: returns are autocorrelated
- ▣ **(B)** Do not reject EMH: the test is inconclusive
- ▣ **(C)** Reject strong-form EMH
- ▣ **(D)** Cannot decide without the variance ratio



## Question 1: Answer

Answer: (A)

- ▣ Since  $Q(10) = 18.5 > 18.31 = \chi_{10,0.05}^2$ , we reject  $H_0$  at 5%.
- ▣  $H_0$ :  $\rho_1 = \dots = \rho_{10} = 0$  (returns are uncorrelated).
- ▣ Rejection means returns have statistically significant autocorrelation  $\Rightarrow$  challenges weak-form EMH.
- ▣ **Caveat**: statistical significance  $\neq$  economic significance. The autocorrelations may be too small to exploit profitably after transaction costs.
- ▣ (B) is wrong:  $Q > \chi_{\text{crit}}^2 \Rightarrow$  we *do* reject. (C) is wrong: Ljung–Box tests weak-form, not strong-form.

## Question 2

### Question

- ☐ A market has Hurst exponent  $H = 0.65$ . This implies:

### Answer choices

- ☐ (A) Returns are anti-persistent (mean-reverting)
- ☐ (B) Returns are consistent with a random walk
- ☐ (C) Returns are persistent (trending behavior)
- ☐ (D) The market is strong-form efficient

## Question 2: Answer

Answer: (C)

- ▣  $H > 0.5$  indicates persistent behavior: positive returns tend to follow positive returns (and vice versa).
- ▣  $H = 0.5$ : random walk (no memory).  $H < 0.5$ : anti-persistent (mean-reverting).  $H > 0.5$ : persistent (trending).
- ▣  $H = 0.65$  is well above 0.5, suggesting significant long-range dependence.
- ▣ This challenges weak-form EMH: past returns contain information about future returns.
- ▣ (A) would require  $H < 0.5$ . (B) requires  $H = 0.5$ . (D) is unrelated to the Hurst exponent.

### Question 3: Reasoning — Event Study Interpretation

#### Question

- ▣ After a positive earnings surprise, a stock's CAR drifts upward by 2% over the following 60 days. This is evidence against:

#### Answer choices

- ▣ (A) Weak-form efficiency only
- ▣ (B) Semi-strong efficiency (the information was public)
- ▣ (C) Strong-form efficiency only
- ▣ (D) No form of efficiency — anomalies are always explained by risk

### Question 3: Answer

Answer: (B)

- ▣ Earnings announcements are **public information**. If the market is semi-strong efficient, prices should adjust immediately and completely.
- ▣ A 60-day drift after the announcement means the market **underreacts** to public information  $\Rightarrow$  violation of semi-strong EMH.
- ▣ This is the Post-Earnings Announcement Drift (PEAD), one of the most robust anomalies.
- ▣ (A) is incomplete: PEAD uses public info, not just past prices. (C) is wrong: insider info is not involved.
- ▣ (D) is Fama's counter-argument, but the drift pattern is systematic and one-directional.

## Exercise 1: Ljung–Box Test and Weak-Form EMH

### Problem

□ A researcher computes the following sample autocorrelations from  $T = 500$  daily stock returns:

| Lag $k$        | 1    | 2     | 3    | 4    | 5     |
|----------------|------|-------|------|------|-------|
| $\hat{\rho}_k$ | 0.08 | -0.03 | 0.05 | 0.02 | -0.01 |

1. Compute the Ljung–Box statistic:  $Q(5) = T(T+2) \sum_{k=1}^5 \frac{\hat{\rho}_k^2}{T-k}$
2. Compare  $Q(5)$  with  $\chi_5^2(0.05) = 11.07$ . Is weak-form EMH rejected?
3. Is  $\hat{\rho}_1 = 0.08$  individually significant? (Use  $z_{\alpha/2}/\sqrt{T}$  with  $z_{0.025} = 1.96$ )
4. Discuss: if  $\hat{\rho}_1$  is significant but  $Q(5)$  is not, what does this mean economically?

## Exercise 2: Variance Ratio Test

### Problem

□ For a stock index, you estimate:  $\hat{\sigma}^2(r_t) = 0.0001$  (daily variance) and  $\hat{\sigma}^2(r_t(5)) = 0.00058$  (5-day return variance), with  $T = 1,000$  observations.

1. Compute  $VR(5) = \hat{\sigma}^2(r_t(5))/(5 \cdot \hat{\sigma}^2(r_t))$
2. Is  $VR(5) > 1$  or  $< 1$ ? Interpret in terms of autocorrelation (momentum vs. mean reversion)
3. Compute the asymptotic variance under homoscedasticity:  $\phi(5) = \frac{2(2 \cdot 5 - 1)(5 - 1)}{3 \cdot 5 \cdot T}$
4. Compute the z-statistic:  $z = (VR(5) - 1)/\sqrt{\phi(5)}$ . Is the random walk rejected at 5%?

### Hint

$z_{0.025} = 1.96$ . Reject  $H_0$  (random walk) if  $|z| > 1.96$ .

## Exercise 3: Event Study

### Problem

- An event study examines  $N = 30$  firms around earnings announcements. Market model parameters:  $\hat{\alpha} = 0.001$ ,  $\hat{\beta} = 1.2$ . On the event day:

|                         | Firm return $R_i$ | Market return $R_m$ |
|-------------------------|-------------------|---------------------|
| Average across 30 firms | 2.5%              | 0.3%                |

1. Compute the average abnormal return:  $\overline{AR} = \bar{R}_i - (\hat{\alpha} + \hat{\beta}\bar{R}_m)$
2. If the standard deviation of  $AR_i$  across firms is  $\hat{\sigma}_{AR} = 1.8\%$ , compute the  $t$ -statistic:  $t = \overline{AR}/(\hat{\sigma}_{AR}/\sqrt{N})$
3. Is the abnormal return significant at 5% ( $t_{0.025, 29} \approx 2.045$ )?
4. If the  $\overline{CAR}(-10, +10) = 3.2\%$  with  $\hat{\sigma}_{CAR} = 4.5\%$ , is the cumulative effect significant?



## Exercise 4: ADF Test Interpretation

### Problem

- A researcher applies the ADF test (with constant, no trend) to daily log-prices of a stock index with  $T = 2,000$  observations. The results are:

|                               |         |
|-------------------------------|---------|
| ADF statistic (prices)        | -2.15   |
| ADF statistic (returns)       | -38.7   |
| Critical value (5%, no trend) | -2.86   |
| Optimal lag (BIC)             | $p = 3$ |

1. Is there a unit root in log-prices? Compare the test statistic with the critical value.
2. Are log-returns stationary? Interpret the ADF on returns.
3. What does this jointly imply about weak-form EMH?
4. If the KPSS test on prices gives statistic = 0.52 with 5% critical value = 0.146, does this confirm or contradict the ADF result?

## Exercise 5: Filter Rule Profitability

### Problem

- A stock has the following daily prices over 10 days: 100, 101.5, 100.8, 99.5, 100.2, 101.8, 103.0, 102.1, 100.5, 101.0. Transaction cost is 0.2% per trade (round-trip 0.4%).
1. Apply a 1% filter rule: buy when price rises 1% from a trough; sell when it falls 1% from a peak. List all trades.
  2. Compute the total return from the filter rule (before and after costs).
  3. Compute the buy-and-hold return over the same period.
  4. Does the filter rule outperform buy-and-hold? What does this imply for weak-form EMH?

### Hint

Track running min (trough) and running max (peak). Buy signal:  $P_t/\text{trough} \geq 1.01$ . Sell signal:  $P_t/\text{peak} \leq 0.99$ .

## Exercise 6: Hurst Exponent Interpretation

### Problem

Four markets have the following estimated Hurst exponents (via R/S analysis on daily returns):

| Market   | $\hat{H}$ |
|----------|-----------|
| Market A | 0.48      |
| Market B | 0.52      |
| Market C | 0.61      |
| Market D | 0.55      |

1. Rank these markets from most efficient to least efficient.
2. Which market(s) exhibit mean reversion? Which exhibit persistence?
3. For Market C with  $H = 0.61$ , what type of trading strategy might be profitable (momentum or mean reversion)?
4. If Market A is an FX pair and Market C is an emerging equity index, is this pattern expected? Why?

## Exercise 7: Event Study Interpretation

### Problem

□ An event study of  $N = 50$  dividend increase announcements yields:

| Window      | $\overline{CAR}$ | $t$ -stat |
|-------------|------------------|-----------|
| $(-1, 0)$   | +1.8%            | 4.2       |
| $(+1, +5)$  | +0.9%            | 1.6       |
| $(+1, +30)$ | +2.5%            | 2.8       |
| $(+1, +60)$ | +3.1%            | 2.1       |

1. Is the immediate reaction  $\overline{CAR}(-1, 0)$  significant at 5%? ( $t_{0.025, 49} \approx 2.01$ )
2. Is the 5-day post-event drift significant? What about the 30-day drift?
3. Does semi-strong EMH hold? Justify using the CAR pattern.
4. Propose a trading strategy to exploit this pattern. What are the risks?

## AI Exercise 2: Behavioral Finance Analysis

Prompt to test in ChatGPT / Claude / Copilot

“Using Python, simulate a portfolio of 20 stocks over 1 year with daily returns drawn from  $\mathcal{N}(0.0003, 0.02^2)$ . Implement an investor who follows the disposition effect: sells any stock that has gained 5% from purchase price, but holds any stock that has lost more than 5%. Compare this disposition-effect portfolio’s terminal value with a buy-and-hold portfolio. Run 1000 simulations and report the average underperformance.”

### Verification checklist:

1. Does the AI correctly track each stock’s purchase price (reference point)?
2. Does it implement the asymmetric selling rule correctly (sell winners at +5%, hold losers below −5%)?
3. After selling a winner, does the AI reinvest the proceeds? In what?
4. Does the simulation account for transaction costs on each sale?
5. Is the comparison fair? (Same random seed, same initial portfolio)
6. Ask: “Now add loss aversion with  $\lambda = 2.25$  to the investor’s utility. How does this change the holding pattern?”

☐ **Warning:** AI may forget to track the reference point after reinvestment, or may apply symmetric

## Quiz: Comprehensive EMH (Questions)

Q1

A market where prices reflect all public and private information is: **(A)** weak-form efficient **(B)** semi-strong efficient **(C)** strong-form efficient **(D)** informationally complete

Q2

The joint hypothesis problem means: **(A)** two assets are always correlated **(B)** efficiency cannot be tested without an equilibrium model **(C)** all anomalies are real **(D)** markets are always efficient

Q3

If  $VR(5) = 0.85$  with  $z = -2.5$ , returns exhibit: **(A)** positive autocorrelation **(B)** negative autocorrelation (mean reversion) **(C)** random walk **(D)** unit root

Q4

Under the AMH, market efficiency is: **(A)** constant across time **(B)** always increasing **(C)** time-varying and regime-dependent **(D)** irrelevant

Q5

The KPSS test differs from ADF because its null hypothesis is: **(A)** unit root **(B)** stationarity **(C)** normality **(D)** no autocorrelation

## Quiz: Comprehensive EMH (Answers)

### Answers

- ▣ **Q1: (C)** — Strong-form efficiency incorporates all information including private/insider information.
- ▣ **Q2: (B)** — Any test of efficiency requires an asset pricing model; rejection may reflect model failure, not inefficiency.
- ▣ **Q3: (B)** —  $VR < 1$  with significant z-statistic indicates negative autocorrelation (mean reversion). Returns that go up tend to go down next.
- ▣ **Q4: (C)** — The AMH posits that efficiency varies with market conditions, investor ecology, and evolutionary pressures.
- ▣ **Q5: (B)** — KPSS reverses the null:  $H_0$  is stationarity, while ADF and PP test  $H_0$ : unit root.
- ▣ **Q6: (B)** — PEAD involves slow adjustment to public information (earnings), which violates semi-strong efficiency.

## Python Lab — Interactive Exercises

### Lab 1: Weak-Form EMH Testing (Y3)

1. Download S&P 500 and BET data (2015–2025); compute log-returns
2. Plot ACF for returns and squared returns
3. Ljung–Box test for lags 10 and 20; interpret  $p$ -values
4. Compute variance ratio for  $q = 2, 5, 10, 20$

### Lab 2: Advanced EMH Analysis (MSc)

1. Compute Hurst exponent via R/S analysis for 5 markets
2. Rolling variance ratio (500-day window) on S&P 500
3. Event study simulation: generate abnormal returns, compute CAR
4. Compare efficiency across developed and emerging markets



## Python: Technical Trading Backtest

```
import yfinance as yf, numpy as np, pandas as pd

sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
p = sp['AdjClose']; r = np.log(p/p.shift(1)).dropna()

# Moving average crossover: MA(50) vs MA(200)
ma50 = p.rolling(50).mean(); ma200 = p.rolling(200).mean()
signal = (ma50 > ma200).astype(int).shift(1) # 1=long, 0=cash
signal = signal.reindex(r.index).fillna(0)

strat_r = signal * r # strategy returns
bh_r = r # buy-and-hold returns

cum_strat = np.exp(strat_r.cumsum()).iloc[-1] - 1
cum_bh = np.exp(bh_r.cumsum()).iloc[-1] - 1
sr_strat = strat_r.mean()/strat_r.std()*np.sqrt(252)
sr_bh = bh_r.mean()/bh_r.std()*np.sqrt(252)

print(f"MA crossover: {cum_strat:.1%}, SR={sr_strat:.2f}")
print(f"Buy & hold: {cum_bh:.1%}, SR={sr_bh:.2f}")
# Typically: MA crossover underperforms buy-and-hold
```

## Python: Sentiment Analysis (VIX vs Future Returns)

```
import yfinance as yf, numpy as np
from scipy import stats

sp = yf.download('^GSPC', start='2010-01-01', end='2025-01-01')
vix = yf.download('^VIX', start='2010-01-01', end='2025-01-01')

# Align dates
common = sp.index.intersection(vix.index)
sp_r = np.log(sp['AdjClose']/sp['AdjClose'].shift(1)).loc[common]

# Forward 20-day return vs current VIX
fwd_20d = sp_r.rolling(20).sum().shift(-20).dropna()
vix_lvl = vix['AdjClose'].loc[fwd_20d.index]

slope, intercept, r_val, p_val, _ = stats.linregress(
    vix_lvl.values.flatten(), fwd_20d.values.flatten())
print(f"Slope: {slope:.4f}, R^2: {r_val**2:.3f}, p: {p_val:.4f}")
# High VIX predicts higher future returns (contrarian signal)
# R^2 is small (~2-5%) but statistically significant
```

## Python: Cross-Asset Rolling Efficiency

```
import yfinance as yf, numpy as np, pandas as pd

tickers = {'^GSPC': 'S&P500', 'GC=F': 'Gold', 'BTC-USD': 'BTC'}
data = {}
for tk, name in tickers.items():
    d = yf.download(tk, start='2018-01-01', end='2025-01-01')
    data[name] = np.log(d['AdjClose']/d['AdjClose'].shift(1))

def rolling_vr(r, q=5, window=500):
    """Rolling variance ratio VR(q)."""
    out = pd.Series(index=r.index, dtype=float)
    for i in range(window, len(r)):
        seg = r.iloc[i-window:i].dropna().values
        var1 = np.var(seg, ddof=1)
        rq = np.array([seg[j:j+q].sum() for j in range(len(seg)-q)])
        varq = np.var(rq, ddof=1)
        out.iloc[i] = varq / (q * var1) if var1 > 0 else np.nan
    return out

# Plot |VR(5) - 1| for each asset on same axes
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(10, 4))
for name, r in data.items():
    vr = rolling_vr(r.dropna())
    ax.plot(vr.index, abs(vr - 1), label=name, alpha=0.7)
ax.legend(); ax.set_ylabel('|VR(5)-1|')
plt.savefig('ch3_rolling_efficiency.pdf', bbox_inches='tight')
```

## Exercise 1: Solution

### Solution

- $$Q(5) = 500 \times 502 \times \left( \frac{0.08^2}{499} + \frac{0.03^2}{498} + \frac{0.05^2}{497} + \frac{0.02^2}{496} + \frac{0.01^2}{495} \right) =$$
$$251,000 \times (0.01282 + 0.00181 + 0.00503 + 0.00081 + 0.00002)/1000$$
$$= 251,000 \times 0.00002049 \approx 251,000 \times 2.049 \times 10^{-5} \approx 5.14$$
- $Q(5) = 5.14 < 11.07 = \chi_5^2(0.05)$ : **do not reject** weak-form EMH
- Individual test:  $|\hat{\rho}_1| = 0.08 > 1.96/\sqrt{500} = 0.0877$ ? No,  $0.08 < 0.0877 \Rightarrow$  **not significant** individually either
- Even if  $\hat{\rho}_1$  were significant, a daily autocorrelation of 0.08 generates expected profit  $\approx 0.08 \times \sigma \approx 0.08\%$  per day — likely consumed by bid-ask spread ( $\sim 0.05\text{--}0.10\%$ )

## Exercise 2: Solution

### Solution

1.  $VR(5) = 0.00058 / (5 \times 0.0001) = 0.00058 / 0.0005 = 1.16$
2.  $VR(5) = 1.16 > 1 \Rightarrow$  positive autocorrelation  $\Rightarrow$  **momentum** (short-term trending)
3.  $\phi(5) = \frac{2 \times 9 \times 4}{3 \times 5 \times 1000} = \frac{72}{15000} = 0.0048$ ;  $\sqrt{\phi(5)} = 0.0693$
4.  $z = (1.16 - 1) / 0.0693 = 0.16 / 0.0693 = 2.31$ . Since  $|z| = 2.31 > 1.96$ : **reject** the random walk at 5%.  
The positive autocorrelation is statistically significant.

### Interpretation

- $VR(5) = 1.16$  implies average autocorrelation  $\bar{\rho} \approx (VR - 1) \times q / (2(q - 1)) = 0.16 \times 5 / 8 = 0.10$ . This is typical for emerging markets and suggests potential inefficiency.

## Exercise 3: Solution

### Solution

1.  $\overline{AR} = 0.025 - (0.001 + 1.2 \times 0.003) = 0.025 - 0.0046 = 0.0204 = 2.04\%$
2.  $t = 0.0204 / (0.018 / \sqrt{30}) = 0.0204 / 0.003286 = 6.21$
3.  $|t| = 6.21 > 2.045 \Rightarrow$  **significant** at 5%. The abnormal return on the event day is highly significant.
4.  $t_{CAR} = 0.032 / (0.045 / \sqrt{30}) = 0.032 / 0.00822 = 3.89 > 2.045 \Rightarrow$  **significant**. The cumulative post-event drift is also significant, suggesting the market underreacts to earnings news (PEAD).

### Discussion

- The fact that  $\overline{CAR}(-10, +10) = 3.2\% > \overline{AR}_0 = 2.04\%$  means the market continued to drift **after** the announcement, violating semi-strong EMH.

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## Key Takeaways — What to Remember from This Chapter

### 5 fundamental lessons

1. EMH is a **joint hypothesis**: efficiency + equilibrium model. Rejection may reflect either.
2. Weak-form tests (autocorrelation, VR, Hurst) generally support efficiency in **developed** markets but reject it in **emerging** markets
3. Event studies reveal robust anomalies (PEAD) that challenge semi-strong efficiency
4. Market efficiency is **not constant**: it varies over time and across markets (AMH)
5. **Statistical significance**  $\neq$  **economic significance**: small autocorrelations may be real but unexploitable after costs

### Preparation for upcoming chapters

- ▣ **Ch. 4:** Extreme Value Theory and Copulas — tail dependence and risk
- ▣ **Ch. 5:** Volatility estimation — GARCH, realized volatility

Thank You! Questions? Course materials available at:

<https://danpele.github.io/SFM/>  Quantlet  Quantinar